Signal Soup: Sharedness vs. code alignment

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This paper introduces *Signal Soup*, a computational model exploring how communication can emerge among multiple agents without assuming they share identical codes. Unlike traditional models that focus on pairs of agents exchanging signals with explicit feedback, *Signal Soup* models stigmergic coordination: agents interact indirectly by modifying and responding to signals in a shared environment. Each agent learns associations between perceived signals and produced responses through simple pattern matching. Simulations reveal two remarkable emergent regularities: agents stabilize at approximately 5.6 sign functions (well below their memory capacity), and the type-token ratio converges to 1/3 regardless of population size. While sign functions become partially shared across agents, the majority remain idiosyncratic. The model suggests that *sharedness*—partial rather than complete code alignment—is sufficient for stable communicative interaction, indicating that loose, distributed alignment may be relevant to robust communication systems.

1. Introduction

Studies on lexical coordination, from Lewis signaling games (Lewis, 1969; Skyrms, 2010) to naming games (Steels, 2011; Scott-Phillips & Kirby, 2010) (Steels, 2011; Scott-Phillips & Kirby, 2010), generally rely on four key assumptions. (1) Arbitrariness: following (Saussure, 1916), there is no dependency (motivatedness) between the two planes (Oliphant, 1996); (2) Direct Feedback: communication success is evaluated through immediate feedback between sender and receiver (Selten & Warglien, 2007); (3) Dyadic Relationship: communication is modeled as occurring between pairs, even in larger groups (Tamariz & Kirby, 2016); (4) Optimal Alignment: ideal communication is defined as complete code sharing among all participants (Steels, 1997). These remarks seem to apply also to recent deep-network agents (Lazaridou & Baroni, 2020). However, these assumptions, though useful for modeling, are quite strong. Though the arbitrariness principle (1) has been challenged by recent research on motivatedness (Dingemanse, Blasi, Lupyan, Christiansen, & Monaghan, 2015), we do not pursue this line here and instead focus on assumptions (2)-(4). Real-world communication is shaped by the environment (2) (Maran, 2020) and often involves multiple participants, suggesting the need for "polylogical" rather than dialogical models (3). Total code sharing (4) is unrealistic: partial alignment usually suffices, as each participant holds an idiosyncratic "encyclopedia" rather than a shared "dictionary" (Eco,

1976, 1986). We introduce *Signal Soup*, a model emphasizing "loose" code alignment, conceived as a *Gedankenexperiment* yet formalized for *in silico* simulation. Signal Soup's dynamics draw on insights from interactive alignment theory (Pickering & Garrod, 2004), which proposes that conversational coordination emerges through automatic priming: encountering an utterance activates representations that become more likely to be produced subsequently. However, while interactive alignment theory models dyadic dialogue with direct feedback, Signal Soup investigates whether similar mechanisms—e.g. priming through repeated exposure, local-to-global alignment, routinization—can operate stigmergically (see later) in polylogical settings without explicit feedback.

2. Signal Soup

In Signal Soup communication is defined as the interaction among communicative agents (CAgs) mediated by a communicative environment (CEnv). CEnv is composed of signals created by CAgs. Each CAg interacts with the environment by processing and replacing signals. Agents and the environment are thus co-dependent. In this sense, agents coordinate through stigmergy (Theraulaz & Bonabeau, 1999): they influence each other indirectly by modifying the shared environment rather than through direct dyadic exchanges. Figure 1 (left) shows signals represented as document shapes; agents process signals. A CAg (Figure

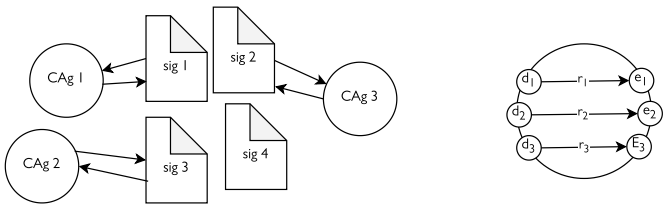

Figure 1. CEnv and CAgs (left), CAg structure (right).

1, right) can retrieve a signal from CEnv and replace it by dispatching a new one. A CAg interacts directly only with CEnv. Signals are generated by assembling the 26 lowercase letters (Kirby & Tamariz, 2021) into three-letter words, yielding $26^3 = 17,576$ possible signals. To retrieve and dispatch a signal, a CAg has an array of detectors and effectors, each defining a three-letter signal type. A sign function (sf) is a mapping $d \rightarrow e$, where d and e denote detector and effector types, respectively. Each sf has an associated integer value representing its relevance r. Relevance has a maximum threshold (by default $r_{max} = 10$), above

which it no longer increases. Each CAg has a fixed memory, defined as the maximum number of sf (by default $sf_{max} = 10$).
A CAg inspects the entire CEnv in search of a signal, with two possible outcomes:

1. If a detector type matches a signal, its corresponding effector is triggered. The incoming signal in CEnv is replaced by a new one generated according to the effector type, and the relevance of sf is increased by 1.
2. If no matching signal is found, the agent selects two random signals from CEnv, links them into a new sf, with an assigned $r = 1$. If sf_{max} is exceeded, a random sf with the lowest relevance is deleted.

A population of CAgs forms a *Band*. During initialization, a Band is randomly assembled with a specified member size n. Each CAg is provided with only one sf, created by coupling randomly assembled signal types. The function is assigned $r = 0$, ensuring that the initial random mappings are easily replaced by the learned ones. All Band members share the same CEnv, which they can access to retrieve and replace signals. CEnv is initially populated with one effector's signal from each agent (thus, it has a default size equal to n, the Band's size). The rationale is that CAgs start populating the CEnv by emitting their only available output signal. The Band operates asynchronously. At each time step (*turn*), one randomly selected CAg inspects the entire CEnv and acts according to the match/no-match procedure described above.

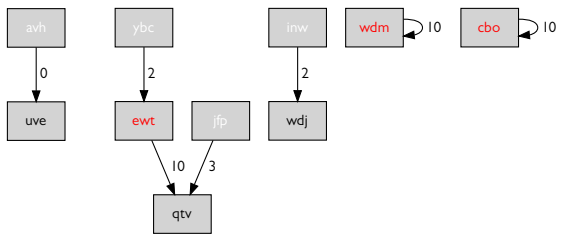

Figure 2. A CAg's topology from a 20-member Band after 2, 000 turns.

3. Results

Figure 2 shows the topology of CAg 0 from a 20-member Band after 2,000 turns. Nodes display signal types: white and black labels represent detectors and effectors, respectively. The CAg is initialized with a single sign function $avh \rightarrow uve$ ($r = 0$), shown on the left. During adaptation, the agent develops a more complex and connected topology. Red labels mark signal types appearing both as detectors

and effectors. A notable emerging feature is looping: types such as wdm and cbo occur as both detectors and effectors. In such cases, the CAg responds to a detected signal by replicating it—effectively responding also to its own behavior. Another pattern involves chain reactions: for instance, ewt is linked to ybc as its effector, while an additional ewt detector connects to the effector qtv, producing a possible longer auto-response sequence from ybc to qtv. Some detectors, such as

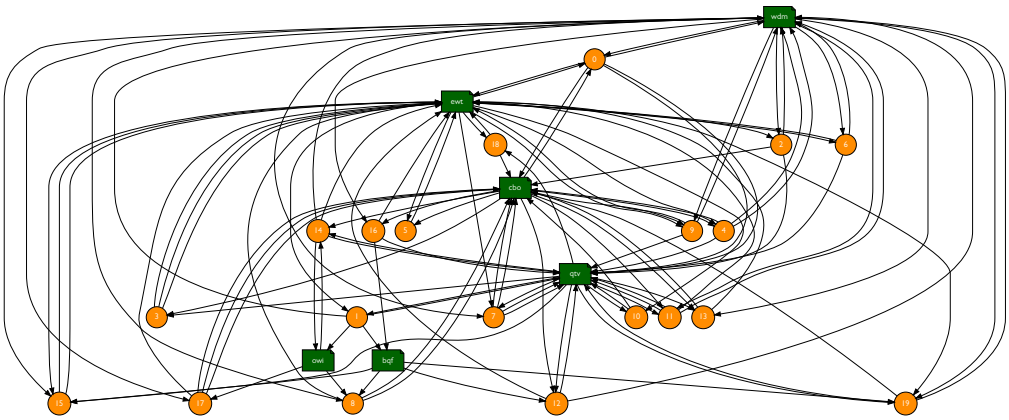

Figure 3. CAgs' matching relations in CEnv for a 20-agent Band after $2,000$ turns.

ewt and jfp, are linked to the same effector (qtv), meaning that different signals elicit the same response—synonymy. Homonymy (a detector linked to multiple effectors) is not excluded per se by the model, rather it is prevented by the interaction dynamics: if a signal matching an existing detector appears in CEnv, the agent responds (case 1) rather than creating a new sf (case 2). Thus, duplicate detectors cannot emerge within an agent's repertoire. Figure 3 then illustrates the relationship between the Band and CEnv. For each signal type in CEnv (green note shapes), if a CAg (orange circle) has a matching detector or effector, a directed edge is drawn to or from that signal type. By design, the actual number of signals in CEnv is $n = 20$, but only six distinct signal types are present. Over time, CEnv's redundancy tends to increase. This trend can be quantified using the type-to-token ratio ($tp : tk$), i.e., the number of different signal types divided by their occurrences (tokens) in the environment. Figure 4 shows the results for Bands ranging in size from 10 to 100 agents. For each size, 100 Bands were created, each run for 10,000 turns. Figure 4, left, shows the average $tp : tk$ ratio. The slope depends on Band size, but in all cases the ratio converges around $1/3$, meaning there are roughly three tokens per signal type. The number of sf

is limited by $sf_{max} = 10$, though this limit is rarely reached, as CAgs typically store fewer sign functions (CAg in Figure 2 stores 7). The average number of sf per agent is shown in Figure 4, right, converging around ~ 5.6. Finally, we aggregate all sign functions across the CAgs in the Band (Figure 5). As before, directed edges represent relations between detectors and effectors, while red edges mark sign functions occurring more than once (see labels). At initialization, the graph consists of 20 disconnected subgraphs $d_i \rightarrow e_i$ (one random sf for each CAg). After $2,000$ turns, the graph develops a central circuit core with red edges, indicating that certain sf are shared among multiple CAgs. In this case, after 2,000 turns, approximately 26% (15 of 56) of sign functions were shared by 2+ agents, with some shared by up to 7 agents, demonstrating emergence of partial alignment without centralized coordination. For example, CAgs 2, 10, and 15 share $ewt \rightarrow wdj$, while CAg 2 shares $wdm \rightarrow cbo$ with CAgs 8 and 17. This level of sharedness is consistent with partial alignment rather than full consensus. In this sense, priming-based coordination can produce functional communication without complete code convergence (Pickering & Garrod, 2004).

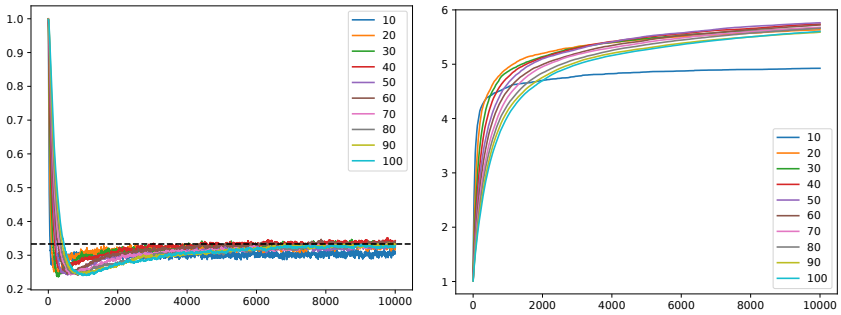

Figure 4. Type/token ratio in CEnv (left) and average sf number per agent (right) for 10 band sizes, 100 bands each, 10,000 turns.

4. Discussion

Signal Soup suggests that alignment can emerge without consensus. Sign functions become partially shared across agents, indicating convergent interaction patterns (alignment in Pickering & Garrod's sense (Pickering & Garrod, 2004)), while the majority remain idiosyncratic. This contrasts with naming games where success is defined as complete code sharing (consensus). Our results suggest that partial code overlap, rather than convergence to identical codes, is sufficient for stable stigmergic coordination. CAgs are modeled after Holland's biologically inspired adaptive agents (Holland, 2014), assuming an ecological interaction with a

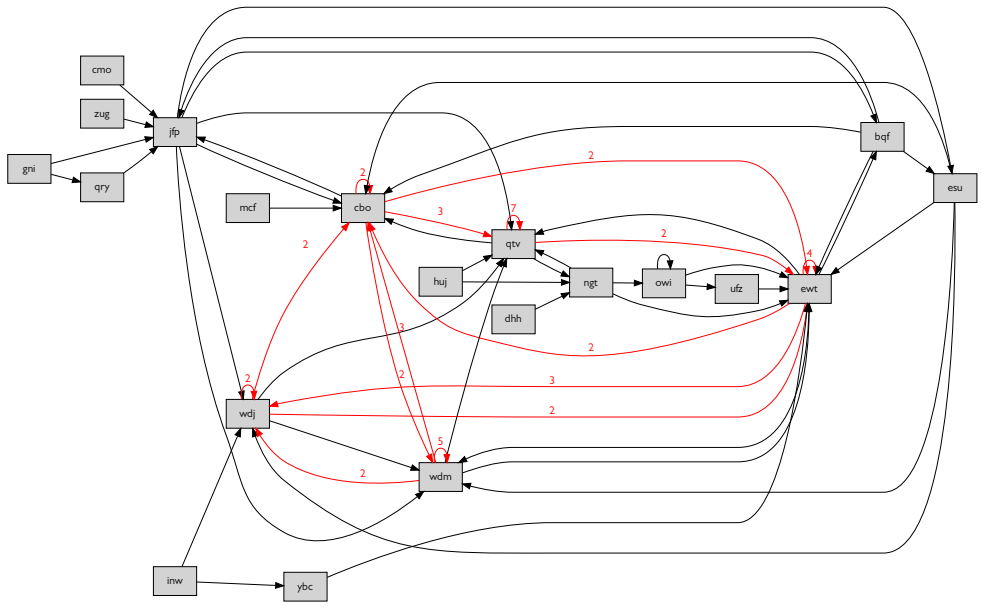

Figure 5. Band topology after $2,000$ turns created by exploding agents. sf with $r = 0$ (initialization) are omitted.

shared communicative environment. The focus here is pragmatic: sign functions act as "rules" prescribing the action to perform when a certain situation occurs (Holland, 1995). There is no symbolic grounding (Steels, 2002); the model emphasizes inter-agent interaction (Hurford, 2007). It must be observed that, after initialization, agents can only recombine signals already in circulation. In this sense, Signal Soup does not model innovation, rather it suggests how a given signal landscape can be effectively refined and shared among agents. The detector-effector edges represent shortcuts to more complex topologies that emerge during simulation (see Figures 2 and 5). Detection and action are mandatory, embodying *Mitteilungsbedürfnis* (Fitch, 2010), akin to a form of enforced Gricean cooperation (Grice, 1989). The output signal replaces the input, modeling communication as sequential, where recent "moves" outweigh history. This aspect also captures the fading nature of signals in oral communication. CEnv's fixed size equals the Band's size, reflecting the sum of agents' communicative actions. CAgs collectively explore CEnv rather than addressing single signals in isolation. When associating two signals into a new sign function, an agent implicitly assumes that others have acted similarly—a minimal theory of mind (cf. the Ob-

verter model (Oliphant & Batali, 1997)). The Band captures the idea of a small, tightly integrated population sharing a communicative environment (Fitch, 2010): it acts as a social unit, forming a core of interconnected sign functions. Increasing graph connectivity and circuit formation (Steels, 1997) recalls autocatalytic sets (Kauffman, 2000, 2019), with looping sign functions yielding "copier" behavior (Fontana, 1991; Kauffman, 1995). The system is also capable of generating longer cycles, where signals circulate through multiple sign functions. Moderate synonymy, rather than being undesirable (Steels, 2011; Tamariz & Kirby, 2016), appears to enhance network connectivity by providing redundant yet reinforcing links inside and among agents. Two emergent properties characterize the system's equilibrium: agents stabilize at ~ 5.6 sign functions (well below the maximum of 10), and the type-token ratio converges to approximately $1/3$ across all band sizes tested—representing an average of 3 tokens per signal type in CEnv, consistent regardless of population size ($n = 10$ to 100 agents). These convergences represent a statistical equilibrium between diversity and redundancy, arising from three interacting factors. First, *reinforcement dynamics*: signals appearing frequently in CEnv are more likely to match detectors, triggering responses that reproduce those same signals, creating positive feedback loops favoring common signals. Second, *emergent repertoire size*: the convergence to ~ 5.6 sign functions suggests this is sufficient for the stigmergic coordination that emerges, with signals appearing repeatedly in CEnv getting reinforced while rare signals are replaced. Third, *stigmergic coordination*: when Agent A produces signal *xyz*, it increases the probability that Agent B will create a sign function incorporating *xyz* (if no match occurs). Importantly, these ratios are not programmed but emerge from the interaction dynamics, and prove robust across parameter variations including memory capacity, population size, and environment size. Unlike these stable equilibria, sharedness levels vary dynamically across runs and band sizes, suggesting a stochastic rather than equilibrium property—the system exhibits dual dynamics where statistical equilibria coexist with continued stochastic variation in which specific sign functions become shared.

5. Conclusions

Signal Soup suggests that partial code overlap is sufficient for communicative coordination. This *sharedness*, or loose code alignment, arises spontaneously from stigmergic priming dynamics without centralized coordination or explicit feedback. The model reveals two remarkable emergent constants: agents stabilize at ~ 5.6 sign functions (well below maximum capacity), and the type-token ratio converges to $\sim 1/3$ regardless of population size or parameter variations. These equilibria represent a balance between diversity and redundancy sufficient for sustained stigmergic coordination. This emergent variability reflects an "encyclopedic" competence in Eco's sense (Eco, 1986), where meanings are distributed and partially shared rather than universally fixed.

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